

NARENDHIRAN SARAVANANE

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Result-driven Robotics Software & Machine Learning Engineer | 2+ yrs exp in ROS2, Docker, Controls & ML-AI fine-tuning | Seeking immediate Contract/FT roles | Open to relocation | Sponsorship required

EDUCATION

MS in Robotics and Autonomous Systems (Honors), Arizona State University | AZ, USA GPA: **4.0 / 4.0** May 2024
B.Tech in Mechanical Engineering, Indian Institute of Technology Patna | Bihar, India GPA: **7.5 / 10** Aug 2022

Tech Finalists: International Robotic Competition (eYRC) & Bachelor's Capstone Project in Mechanical Department
Conferences: IEEE, American Control Conference ACC'24 | Indian Institute of Science, I-4AM'22 | Delivered 2 talks
Courses: Linear Algebra | Sequential Decision Making | RL | ML | UAVs | Perception | Optimization | Controls | PDE

EXPERIENCE

DreamFace Technologies, Robotics Software Engineer | Denver, USA Mar 2025 - Present

- Orchestrated **behavior engine** managing LLM-driven decisions via **lifecycle nodes**, modular cum scalable actions
- Designed **micro-ROS** framework on Teensy 4.0 microcontroller, established ZLAC motor control via **RS485** protocol
- Engineered automated health monitoring systems with per-node diagnostics, elevated system **reliability by 10x**
- Configured Dynamixel & Feetech motors in **daisy-chain** topology for humanoid, integrated ros2_control framework

Padma Agrobotics, Robotics Software Engineer | Arizona, USA May 2024 - Oct 2024

- Trained **Yolo** algorithm with field-collected datasets to detect new features and fine-tune model parameters
- Architect **Docker-ROS2** nodes, and establish GPS, camera, sensors & pneumatics through **ethernet & digital IO**
- Developed automated data processing pipelines, parsing & logging **ros2bag** binary data for large-scale analysis
- Configured motor systems, motor controllers, and joystick operations via **PLC** to apply dynamic control algorithms

Brainchip, Solutions Architect Intern (Robotics & RL Specialist) | Remote (California, USA) May 2023 - Aug 2023

- Constructed **3D models** and ROS-joints controlled through Q-Learning, **RL** model operated on AKD1000 Chip
- Spearheaded the end-to-end creation of an AI-controlled robot, accelerating project completion by **40%**
- Transformed the AI model's transition from TF to BrainChip's MetaTF **framework**, enriching customer acquisition

Indian Institute of Technology Bombay, Robotic Software Engineer Intern | Remote (India) May 2020 - Aug 2020

- Led an 4-person team to develop a fiducial-marker-based **localization** model for an unstable camera feed
- Achieved high-precision localization model in V-rep for real-time camera feeds, reducing calibration error to $\leq 0.5\%$
- Amplified design, combined rule-based script, and **unit tested** to validate auto-evaluators with 95% coverage

SKILLS

Languages Python, C/C++, embedded C, Catkin, CUDA, CMake, Matlab, Git, Bash, LaTeX, Vim, PCL, I2C
Robotics ROS 1/2, V-Rep, Gazebo, MoveIt, Arduino, AtMega 2560, Sensor Fusion, PLC, ASR, Whisper
Software Linux, Tensorflow, Pytorch, Docker, OpenCV, ZeroMQ, B0RemoteAPI, CorelDraw, Solidworks
Certifications Robotics Software Engineer, Udacity Nanodegree – (2023) | Self-Driving Cars, University of Toronto – (2023)

PROJECTS

Cup Stacking Robot – Embodied AI Hackathon (Seed Studio & NVIDIA) Oct 2025

- Fine-tuned **foundational models** NVIDIA GR00T 1.5 and **ACT** for autonomous cup stacking on SO-101 robotic arm
- Collected **80+ teleoperated** demonstrations, trained imitation learning policy for grasp-transfer-stack sequence
- Deployed model inference on **Jetson Thor** edge AI platform, achieved robust real-time control performance

Home-Delivery Bot | Robotics Software Engineer, Udacity Nanodegree (Scholarship Scholar) Dec 2022 - April 2023

- Developed a robot in Gazebo (**ROS**) & integrated with **feedback control** for state dynamics
- Implemented SLAM and sensor fusion (Rotary Encoder, Odom & IMU) for navigation & deployed AMCL

IEEE Paper – Control Systems Society Conference (Paper Published) Jan 2023 - Jan 2024

- Distributed RHC approach for multi-agent systems with privacy and maintained MTL specifications
- Utilized Kalman filter equations and **MILP** to encode causal MTL specifications as constraints

Visual Tracking UAV - Mambo Drone Jan 2023 - April 2023

- Tailored a high-performance, **low-level flight control** algorithm with an integrated **Kalman Filter** for an Drone
- Championed an optimized red color detection algorithm, slashed processing time by 30%, and improved efficiency.

Meta's Research Enhancement – Object Goal Navigation Jan 2023 - April 2023

- Integrated **YOLOv7** and performance enhancements led to a 7% success rate boost in object goal navigation
- Engaged with a deep RL model, leveraged On Policy. Integrated **RRT** to path planning replaced Fast Marching

Dc-GANs (Deep Convolutional Generative Adversarial Network) – Fashion MNIST Nov 2022 - Dec 2022

- Devised a DcGAN **neural** architecture & successfully trained within 50 epochs to generate realistic synthetic images
- Crafted a Tensorflow-based neural network and optimized the generator and discriminator model